Hints: Differentiable functions – 1

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Substitute $t = nx$: $\displaystyle\int_0^1 f(nx)\,dx = \frac{1}{n}\int_0^n f(t)\,dt$, which is an average of f over $[0, n]$.

Hint 2. Apply l'Hôpital to $F(y)/y$ with $F(y) = \int_0^y f$ (the denominator tends to ∞, $F' = f \to L$). Alternatively, split $[0, n]$ into $[0, A]$, where f is bounded, and $[A, n]$, where f is close to L (or larger than any given M if $L = +\infty$).

Problem 2

Hint 1. Fix x with $a = f'(x) \neq 0$; by symmetry ($x \mapsto -x$) assume $a > 0$. Use positivity of f at a point to the left of x.

Hint 2. The Lipschitz condition gives $f'(x - t) \geqslant a - Lt$ for $t \geqslant 0$. Integrate over $t \in [0, a/L]$: $f(x - a/L) \leqslant f(x) - \int_0^{a/L}(a - Lt)\,dt = f(x) - \dfrac{a^2}{2L}$.

Hint 3. Since $f(x - a/L) > 0$, this gives $a^2 < 2Lf(x)$. The case $a = 0$ is trivial because $f > 0$.

Problem 3

Hint 1. Factor the operator: $f'' - 5f' + 6f = (D - 3)(D - 2)f$, and note that $3e^{2x} - 2e^{3x}$ is the solution of the equation with the same initial data.

Hint 2. Put $g = f' - 2f$. Then $g' - 3g \geqslant 0$, i.e. $(ge^{-3x})' \geqslant 0$, and $g(0) = -2$, so $g(x) \geqslant -2e^{3x}$ for $x \geqslant 0$.

Hint 3. Now $(fe^{-2x})' = ge^{-2x} \geqslant -2e^{x}$. Integrate from 0: $f(x)e^{-2x} - 1 \geqslant -2(e^x - 1)$, which is exactly the claim.

Problem 4

Hint 1. Guess first: for small x both are $x + \frac{x^3}{6} + \dots$, so expand further, or test values near $\frac{\pi}{2}$. The answer is $\tan(\sin x) > \sin(\tan x)$ for all $x \in (0, \frac{\pi}{2})$.

Hint 2. If $\tan x \geqslant \frac{\pi}{2}$, i.e. $x \geqslant \arctan\frac{\pi}{2}$, then $\sin(\tan x) \leqslant 1$, while $\sin x \geqslant \sin\arctan\frac{\pi}{2} = \frac{\pi}{\sqrt{4+\pi^2}} > \frac{\pi}{4}$, so $\tan(\sin x) > 1$.

Hint 3. On $(0, \arctan\frac{\pi}{2})$ let $h(x) = \tan(\sin x) - \sin(\tan x)$, $h(0) = 0$. Then $h' > 0$ is equivalent to $\cos(\tan x)\cos^2(\sin x) < \cos^3 x$.

Hint 4. Since $\ln\cos$ is concave on $(0, \frac{\pi}{2})$, $\cos(\tan x)\cos^2(\sin x) \leqslant \cos^3\frac{\tan x + 2\sin x}{3}$. Finally $\tan x + 2\sin x > 3x$ (differentiate: $\sec^2 x + \cos x + \cos x \geqslant 3$ by AM–GM), and cos is decreasing.

Problem 5

Hint 1. Look for a substitution $g = \varphi f$ that turns $f'' + 2xf' + (x^2 + 1)f$ into $\varphi\, g''$ times something simple.

Hint 2. With $g = e^{x^2/2}f$ one gets $g'' = e^{x^2/2}\big(f'' + 2xf' + (x^2 + 1)f\big)$, so $|g''(x)| \leqslant e^{x^2/2}$.

Hint 3. Apply l'Hôpital twice to $f = g/e^{x^2/2}$ (the denominators $e^{x^2/2}$ and $xe^{x^2/2}$ tend to ∞): the last ratio is $\dfrac{g''}{(1+x^2)e^{x^2/2}}$, of absolute value $\leqslant \frac{1}{1+x^2} \to 0$.

Problem 6

Hint 1. The answer is no. What does $f' = f \circ f > 0$ say about monotonicity, and then about $f(f(x))$?

Hint 2. f is strictly increasing and $f(x) > 0$, so $f'(x) = f(f(x)) > f(0) =: c > 0$ for every x.

Hint 3. Then $f(x) \leqslant f(0) + cx$ for $x < 0$, which becomes negative for x very negative. Contradiction.

Problem 7

Hint 1. Key observation: if $f(s) > 0$, then at every point r with $f(r) = s$ we have $f'(r) > f(f(r)) = f(s) > 0$. Show that f then takes the value s at most once, and if it does, at r say, then $f < s$ on $(-\infty, r)$ and $f > s$ on (r, ∞).

Hint 2. Second lemma: $f(x) \to +\infty$ as $x \to +\infty$ is impossible. Otherwise $f' \to \infty$, f is eventually increasing with $f(x) \geqslant x + 1$, and then $f(x+1) - f(x) = f'(\xi) > f(f(\xi)) \geqslant f(x+1)$, so $f(x) < 0$. In particular $f' > c > 0$ on a half-line is impossible.

Hint 3. Now suppose $t \geqslant 0$, $v = f(t)$, $u = f(v)$ and $f(u) > 0$. Apply Hint 1 to the level u: its only preimage is v, with $f < u$ to the left of v and $f > u$ to the right. If $u \leqslant 0$, compare u with v: $f(u) > 0 \geqslant u$ forces $u > v$, but then $t \geqslant 0 \geqslant u > v$ gives $f(t) > u$, i.e. $v > u$. Contradiction.

Hint 4. If $u > 0$: when $u \geqslant v$, we get $f > u$ on (v, ∞), hence (as $f(x) > u \geqslant v$) $f'(x) > f(f(x)) > u$ for $x > v$, contradicting Hint 2. When $u < v$, then $f(u) < u$; now $v > 0$ and $f(v) = u > 0$, so apply Hint 1 to the level v: $f > v$ on (t, ∞), hence $f(f(x)) > u$ and $f'(x) > u$ there. Again this contradicts Hint 2.

Problem 8

Hint 1. A 2-cycle is a fixed point of $g = f \circ f$ that is not a fixed point of f. Compute $g'(\xi)$ with the chain rule (it only needs differentiability of f at $\xi = f(\xi)$).

Hint 2. $g'(\xi) = f'(\xi)^2 > 1$, so $g(x) < x$ for $x < \xi$ close to ξ. Also $f'(\xi) < 0$ gives $f(x) > \xi$ just to the left of ξ.

Hint 3. Let a be the left end of the maximal interval (a, ξ) on which $f > \xi$. Then either $f(a) = \xi$, so that $g(a) = \xi > a$, or $a = 0$ and $g(0) \geqslant 0$. By the IVT, $g(x) - x$ vanishes at some $x_0 \in [a, \xi)$ with $f(x_0) > \xi$ (check the endpoint cases). Put $x_1 = f(x_0)$.

Problem 9

Hint 1. Argue by contradiction: if the product is negative everywhere, then f, f', f'', f''' are all nonvanishing, so each has a constant sign.

Hint 2. Replacing $f(x)$ by $-f(x)$ or by $f(-x)$ does not change the sign of the product, so you may assume $f > 0$ and $f' > 0$. Then f'' and f''' have opposite signs.

Hint 3. If $f'' > 0 > f'''$, then $f'' \geqslant f''(0) > 0$ on $(-\infty, 0]$, so $f' \to -\infty$ as $x \to -\infty$. If $f'' < 0 < f'''$, then $f'' \leqslant f''(0) < 0$ on $(-\infty, 0]$, so $f' \to +\infty$ as $x \to -\infty$ and hence $f \to -\infty$. Both contradict the signs.

Problem 10

Hint 1. First study $x \to -\infty$: f, f', f'' are positive and increasing, so they have limits there. Show that $f' \to 0$ and $f'' \to 0$.

Hint 2. Multiply $f''' \leqslant f$ by $f'' > 0$ and integrate from $-\infty$: $\frac{1}{2} f''^2 \leqslant \int_{-\infty}^{x} f f'' = f f' - \int_{-\infty}^{x} f'^2 \leqslant f f'$. So $f'' \leqslant \sqrt{2 f f'}$.

Hint 3. Then $\left(f'^{3/2}\right)' = \frac{3}{2} f'^{1/2} f'' \leqslant \frac{3}{\sqrt{2}} f' \sqrt{f} = \sqrt{2}\left(f^{3/2}\right)'$. Integrating from $-\infty$ gives $f' \leqslant 2^{1/3} f < 2f$.